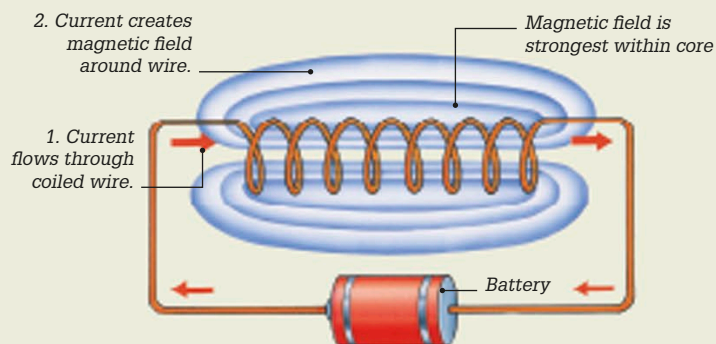


1820 ELECTROMAGNETISM

Danish physicist Hans Christian Ørsted found that a wire carrying an electric current made a magnetized compass needle move. This inspired French physicist André-Marie Ampère to experiment further with magnets and electricity, and produce a theory of electromagnetism.



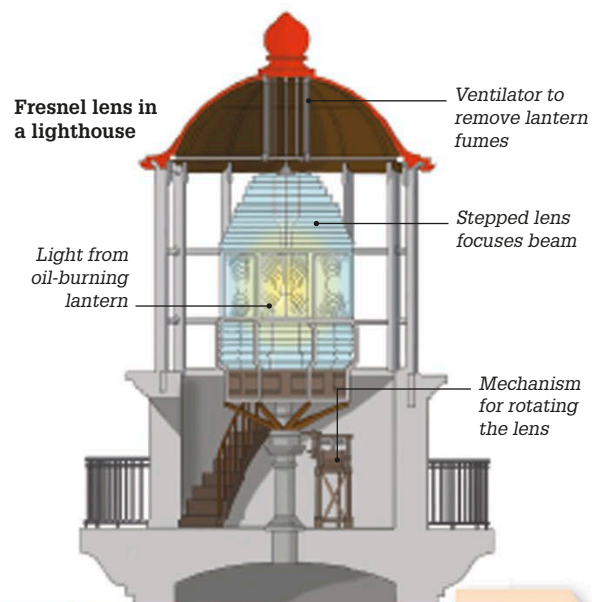
Electromagnet

In 1822, Ampère discovered that a coil of wire carrying an electric current produces a magnetic field just like that of a bar magnet. Adding an iron bar inside the coil intensifies the effect, and a switch allows it to be turned on and off. In 1829, US scientist Joseph Henry created some very powerful electromagnets by winding the coils more tightly and adding several layers of coils.

1823

Light at sea

French scientist Augustin-Jean Fresnel invented a special lens to be used in lighthouses. The Fresnel lens, made up of stepped concentric circles, concentrated light into a powerful narrow beam. Ships up to 20 miles (32 km) away could see the light, preventing them from running aground or crashing onto rocks.



1825

First steam crossing

The SS *Savannah* was the first ship driven partly by a steam engine to cross the Atlantic Ocean. She left Savannah, Georgia, on May 22, 1819 arriving in Liverpool, England, 18 days later.



Postage stamp showing SS *Savannah*

The story of engines
See pages 130–131

Creature stood 9 ft (3 m) high

Sharp, jagged teeth for eating prey

1822

Hard to stomach

US Army surgeon William Beaumont studied digestion by dangling pieces of food into the stomach of a patient with an open wound (after a gunshot accident) and then pulling them out to see how the gastric juices were working.

1824

Giant lizard

British naturalist William Buckland identified several fossil bones as those of an extinct reptile. He called it *Megalosaurus* (giant lizard). It was the first scientific description of a dinosaur.

Reconstructed skeleton of a *Megalosaurus*

British geologist Richard Owen invented the word “dinosaur” (terrible lizard) in 1842.